Status	Finished
Started	Tuesday, 13 January 2026, 8:30 AM
Completed	Tuesday, 13 January 2026, 9:19 AM
Duration	49 mins 2 secs

Question 1

Complete

There are 10 men and 8 women in an organization. To host an activity, a committee of 6 members is required. If the committee must include at least 2 men and at least 3 women, how many different committees can be formed?

- 9870
- 44520
- 110880
- none of these
- 6720

Question 2

Complete

If T is a tree and includes all the vertices of a connected graph G, then T is a ____ tree of G.

- a. Maximum Spanning
- b. Spanning
- c. Minimal
- d. Minimum Spanning
- e. Fundamental

Question 3

Complete

In graph algorithms, Dijkstra's algorithm is usually used to:

- a. Find shortest paths from one source to all other vertices in a weighted graph with non-negative edge weights
- b. Find a minimum spanning tree
- c. Find the shortest path between two vertices in any graph
- d. Sort vertices in non-decreasing order of degree
- e. None of these

Question 4

Complete

Let f(x) = 2x + 1 with Domain A = {1,2,3,4} and Codomain B = {1,2,3,4,5,6,7,8,9,10}.

What is the Range (image) of f on A?

- {3,5,7,9}
- {1,2,3,4,5,6,7,8,9,10}
- {2,4,6,8}
- {1,2,3,4}
- None of these

Question 5

Complete

Given $f(x) = 2x + 3$ and $g(x) = x^2$, what is $(g \circ f)(x)$?

- $g \circ f(x) = 2x + 9$
- $g \circ f(x) = (2x + 3)^2$
- $g \circ f(x) = 2x^2 + 3$
- $g \circ f(x) = x^2 + 6$
- None of these

Question 6

Complete

If $f(x) = 2x + 3$, what is the inverse function $f^{-1}(x)$?

- $f^{-1}(x) = (x + 3)/2$
- None of these
- $f^{-1}(x) = 2x - 3$
- $f^{-1}(x) = x^2 - 3$
- $f^{-1}(x) = (x - 3)/2$

Question 7

Complete

Let A={1,2,3,4}, B={a,b,c,d}, C={x,y,z}

R = { (1,a) , (2,d) , (3,a) , (3,b), (3,d) }

S = {(b,x), (b,z) , (c,y) , (d,z)}

What is S o R ?

- {(2,x), (3,y), (4,z)}
- {(2,z), (3,x), (3,z)}
- {(1,x), (1,z), (2,y)}
- {(1,z), (2,z), (4,y)}
- None of these

Question 8

Complete

For matrices A (size 2×3) and B (size 2×5), what is the size of the product BA?

- A. None of these
- B. Multiplication is not possible
- C. 2×2
- D. 5×3
- E. 3×5

Question 9

Complete

How many distinct 10-letter arrangements can be made from the word TERENGGANU if the two 'A's must be not be adjacent ?

- 362880
- 816480
- 3265920
- 272160
- None of these

Question 10

Complete

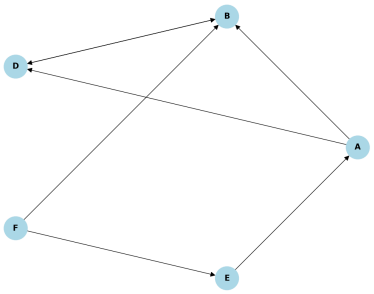
Five students are going to line up to buy lunch in Kafe D'Rimba. How many different orders can the form the line?

- None of these
- 300
- 20
- 60
- 5

Question 11

Complete

Refer to the following diagraph:



Identify the relation presented by the diagraph.

- { (A, B), (B, D), (D, A), (E, F), (F, A), (E,B) }
- { (A, B), (B, D), (A, D), (D, B), (E, F), (F, A), (E,B) }
- None of these
- { (A, B), (B, D), (A, D), (D, B), (F, E), (E, A), (F,B) }
- { (A, B), (B, D), (A, D), (D, B), (E, F), (F, A), (E,B), (C, NULL) }

Question 12

Complete

The determinant of the matrix

$$\begin{pmatrix} 5 & -19 & 1 \\ 9 & -2 & 4 \\ -10 & -4 & 18 \end{pmatrix}$$

is:

- A. 3682
- B. 3522
- C. -3994
- D. -2594
- E. None of these

Question 13

Complete

Given the following system of linear equations:
3x - 5y = 7
2x - 9y = 5
Using the Cramer's method to solve the problem, we need to find the coefficient matrix D, followed by D_x. Identify D_x.

- $\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}$
- $\begin{pmatrix} 3 & 5 \\ 2 & 9 \end{pmatrix}$
- None of these
- $\begin{pmatrix} 7 & -5 \\ 5 & -9 \end{pmatrix}$
- $\begin{pmatrix} 3 & -5 \\ 2 & -9 \end{pmatrix}$

Question 14

Complete

Which operation is NOT defined for matrices in general?

- A. None of these
- B. Scalar multiplication
- C. Addition of two matrices of same size
- D. Division of one matrix by another
- E. Transpose

Question 15

Complete

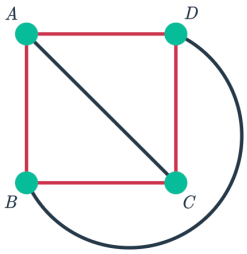
Let G be a simple graph with n vertices. What is the maximum possible number of edges in G?

a. none of these
b. $\frac{n(n-1)}{2}$
c. n^2
d. $\frac{n(n+1)}{2}$
e. 2n

Question 16

Complete

For the graph shown below, which option lists only Hamiltonian cycles?



a. ACDA and DCABD
b. None of these
c. ABDA and BADC
d. ABCDA and ABACD
e. ABDCA and DBACD

Question 17

Complete

A binary tree of depth d (with the root at depth 0) can have at most how many nodes?

a. None of these
b. 2^d-1
c. 2^d
d. d^2
e. 2^{d+1}

Question 18

Complete

Find the determinant of the following matrix:

$$\begin{bmatrix} 3 & 4 & -2 \\ 2 & 5 & 5 \\ 1 & -3 & 6 \end{bmatrix}$$

- 173
- None of these
- 85
- 89
- 129

Question 19

Complete

Which function is NOT one-to-one (injective) on the real numbers R?

- None of these
- f(x) = x - 10
- f(x) = 2x + 3
- $f(x) = x^2 + 9$
- f(x) = 5x + 9

Question 20

Complete

For matrix multiplication AB to be defined, which condition must be satisfied?

- A. Number of columns of A = Number of columns of B
- B. None of these
- C. Number of columns of A = Number of rows of B
- D. Number of rows of A = Number of rows of B
- E. Both matrices must be square

Question 21

Complete

Evaluate the floor function: floor(3.9) =

- -3
- 3
- None of these
- -4
- 4

Question 22

Complete

A function f: X -> Y is surjective (onto) if:

- Different inputs in X always produce different outputs in Y.
- No element in Y is mapped from X.
- Each x in X maps to more than one y in Y.
- None of these
- Every y in Y is hit by at least one x in X (Range = Codomain).

Question 23

Complete

If det(P) = 7 and P is 3×3, what is det(3P)?

- A. 63
- B. 441
- C. None of these
- D. 21
- E. 189

Question 24

Complete

A simple unweighted graph has 7 vertices. How many entries are there in its adjacency matrix?

- a. none of these
- b. 7
- c. 49
- d. 36
- e. 14

Question 25

Complete

Given A and B are matrices. Under which condition the following equation is TRUE?

$A^2 - B^2 = (A-B)(A+B)$

- None of these
- A is a row matrix and B is a column matrix
- B is a unit matrix
- A is a column matrix and B is a row matrix
- AB = BA

Question 26

Complete

Given matrix

$$B = \begin{pmatrix} 3 & 0 & 1 \\ 9 & -2 & 0 \\ -1 & 0 & 1 \end{pmatrix}$$

if det(B) = m, the inverse of B is:

- A. $\frac{1}{m}\begin{pmatrix} -2 & 0 & 2 \\ 9 & 4 & -9 \\ -2 & 0 & -6 \end{pmatrix}$
- B. None of these
- C. $\frac{1}{m}\begin{pmatrix} -2 & -9 & -2 \\ 0 & 4 & 0 \\ 2 & 9 & -6 \end{pmatrix}$
- D. $\frac{1}{m}\begin{pmatrix} -2 & 0 & 2 \\ -9 & 4 & 9 \\ -2 & 0 & -6 \end{pmatrix}$
- E. $\frac{1}{m}\begin{pmatrix} -2 & 9 & -2 \\ 0 & 4 & 0 \\ 2 & -9 & -6 \end{pmatrix}$

Question 27

Complete

Which statement best describes a function f: X -> Y ?

- Every y in Y must be paired with exactly one x in X.
- A function must contain all pairs in X x Y.
- Every x in X is paired with exactly one y in Y.
- None of these
- Some x in X can be paired with two different y in Y.

Question 28

Complete

How many distinct 10-letter arrangements can be made from the word TERENGGANU?

- None of these
- 5040
- 3628800
- 1814400
- 453600

Question 29

Complete

Given the following system of linear equations:
3x - 5y = 7
2x - 9y = 5
Using the Cramer's method to solve the problem, we need to find the coefficient matrix D. Identify D.

- $\begin{pmatrix} 3 & 5 \\ 2 & 9 \end{pmatrix}$
- \(\begin{pmatrix} 7 & -5 \\ 5 & -9 \end{pmatrix} \)
- \(\begin{pmatrix} 3 & -5 \\ 2 & -9 \end{pmatrix} \)
- None of these
- \(\begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix} \)

Question 30

Complete

Given R = {(2,3), (2,5), (3,4), (3,6), (6,6)}, what is R^-1 ?

- {(2,3), (2,5), (3,4), (3,6), (6,6)}
- {(2,2), (3,3), (4,4), (5,5), (6,6)}
- None of these
- {(3,5), (5,3), (4,6), (6,4), (6,6)}
- {(4,3), (6,3), (6,6), (3,2), (5,2)}

Question 31

Complete

Which condition guarantees that a connected undirected graph has an Eulerian circuit?

a. Exactly two vertices have odd degree
b. none of these
c. All vertices have odd degree
d. All vertices have even degree
e. The graph is complete

Question 32

Complete

In a Tree, nodes beneath its parent have a level____ than its own?

a. One Lower
b. One Higher
c. none of these
d. Two Lower
e. Two Higher

Question 33

Complete

If |X| = 3 and |Y| = 5, how many different functions f: X -> Y exist?

- 3^5
- 5!
- 5^3
- 3 * 5
- None of these

Question 34

Complete

Given x, y ∈ A where A is the set of students of UM. If R is a relation such that xRy if x is at least taller than y, R is _____ .

- Not reflexive and symmetric
- symmetric and non transitive
- none of the above
- Reflexive and transitive
- antisymmetric and non transitive

Question 35

Complete

A bag contains marbles of five different colors: red, blue, green, yellow, and white. If you reach into the bag without looking, what is the minimum number of marbles you must pull out to guarantee you have at least three marbles of the same color?

- 15
- 12
- 11
- 13
- None of these

Question 36

Complete

In a rooted tree, what is the level of the root?

a. 0

b. None of these

c. 1

d. 2

e. It depends on the particular tree

Question 37

Complete

In graph theory, which description best matches a tree?

a. Any graph with exactly one path between some pair of vertices

b. A connected undirected graph with no cycles

c. An undirected graph where every vertex has degree 2

d. None of these

e. A connected undirected graph that may contain cycles

Question 38

Complete

A relation R on a set X is an equivalence relation if it is:

Reflexive, symmetric, and transitive

Symmetric and non transitive

Symmetric only

None of these

Antisymmetric and transitive

Question 39

Complete

Let A = {1,2,3} and B = {1,2}. Define relation R from A to B by:

(a,b) is in R if a > b.

Which zero-one matrix M_R is correct, assuming rows in order a=(1,2,3) and columns in order b=(1,2)?

- $$\begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 1 & 1 \end{bmatrix}$$
- $$\begin{bmatrix} 1 & 1 \\ 1 & 0 \\ 0 & 0 \end{bmatrix}$$
- $$\begin{bmatrix} 0 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$
- $$\begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}$$
- None of these

Question 40

Complete

A relation R on set X is reflexive if:

- None of these
- For all a,b,c in X, if (a,b) is in R and (b,c) is in R then (a,c) is in R.
- For all x in X, (x,x) is in R.
- For all x in X, if $x \neq 0$, (x,x) is in R.
- For all a,b in X, if (a,b) is in R then (a,a) is not in R.